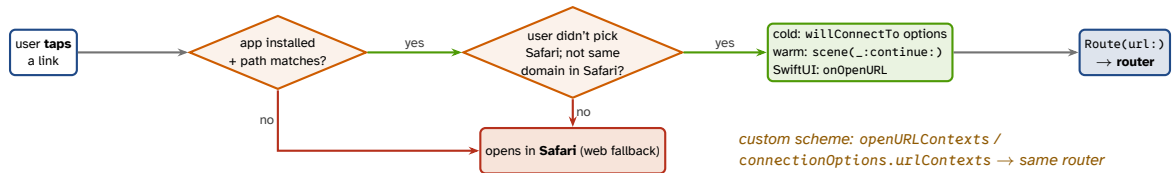


In one line: A **Universal Link** is a plain `https://` URL that opens your app *only if* the domain has proved, via its `apple-app-site-association` file, that it trusts your app ID; otherwise it opens the website. A **custom scheme** (`myapp://`) is unverified, can be claimed by any app, and fails when the app is missing. Every entry point should parse to one route and hand it to one **router**.

1 · Verification (at install / update, then periodically)



2 · A tap on `https://example.com/p/42`



How it works

- AASA:** JSON at `https://<domain>/well-known/apple-app-site-association`: no `.json` extension, `application/json`, valid TLS, **no redirects**, ≤ 128 KB, unsigned. `appIDs` = `TEAMID.bundle.id`. components match `"/` path, `"?"` query, `"#"` fragment, with `*/?` wildcards and `"exclude": true`; first match wins. The same file also serves webcredentials (passwords), appclips and activitycontinuation (Handoff).
- Associated Domains** entitlement: `applinks:example.com`, `applinks:*.example.com`. Each subdomain counts as a separate domain; appclips: takes no wildcard.
- Apple's CDN** (iOS 14+): the device fetches from Apple, not from you; the CDN fetches within **24 h**, devices re-check about **weekly** — AASA changes are **not instant**. In development use `applinks:host?mode=developer` + Settings → Developer → Associated Domains Development.
- Entry points:** a Universal Link is an `NSUserActivity` (`NSUserActivityTypeBrowsingWeb`, `.webpageURL`). **Cold:** `connectionOptions.userActivities` in `willConnectTo`. **Warm:** `scene(_ : continue:)`. Scheme (`CFBundleURLTypes`): `warm scene(_ : openURLContexts:)`, `cold connectionOptions.urlContexts`. `SwiftUI: onOpenURL` gets both; `onContinueUserActivity` also works.
- Routing:** parse once (`URLComponents` → `enum Route`). Scenes, push taps, Quick Actions and Spotlight all call the **same coordinator**. At cold start **hold the route** until the UI and session are ready (e.g. after login).

AASA

```
{ "applinks": { "details": [ {
  "appIDs": [ "ABCDE12345.com.you.app" ],
  "components": [
    { "/": "/help/*", "exclude": true },
    { "/": "/p/*", "?": { "ref": "?*" } }, // needs ?ref=
    { "/": "/order/*" } ] ] ] }
```

Remember

Prove · Tap · Route: the domain proves, the user taps, one coordinator routes.

Example

```
func scene(_ s: UIScene, willConnectTo _: UISceneSession,
           options o: UIScene.ConnectionOptions) {
  setUpWindowAndCoordinator() // first
  if let url = o.userActivities.first?.webpageURL
    ?? o.urlContexts.first?.url { coordinator.open(url) } }
func scene(_ s: UIScene, continue ua: NSUserActivity) {
  guard ua.activityType == NSUserActivityTypeBrowsingWeb,
    let url = ua.webpageURL else { return }
  coordinator.open(url) }
func scene(_ s: UIScene,
           openURLContexts c: Set<UIOpenURLContext>) {
  if let url = c.first?.url { coordinator.open(url) } }
// open: Route(url:) -> show screen; unknown route -> Safari
// SwiftUI: rootView().onOpenURL { router.open($0) }
```

Why did it open Safari?

- AASA invalid or stale: a **redirect** (proxy 301 to `www`), wrong type, bad cert, or the CDN still has the old copy.
- Entitlement or team ID wrong; path not matched, or hit an exclude.
- URL **typed or pasted** into Safari. Only *taps* count.
- Same-domain navigation:** a link on `example.com` to `example.com` stays in Safari.
- The user once chose **Safari** (breadcrumb or long-press). This is sticky per domain; long-press → Open in App undoes it.
- Loaded in your own `WKWebView`, or a JS redirect with no tap.

Universal Link vs custom scheme

	<code>https://</code> Universal Link	<code>myapp://</code> scheme
proves the app	the domain's AASA	nothing — any app can claim
app not installed	opens the website	fails
declared in	Associated Domains	<code>CFBundleURLTypes</code>
arrives as	<code>NSUserActivity</code>	<code>UIOpenURLContext</code>

Traps

- Only handling `scene(_ : continue:)`: cold-start links arrive in `willConnectTo` and get lost.
- Custom schemes are not proof of identity: never put tokens or secrets in them.
- Deferred** deep links (install, then land on the content) are not built in: attribution SDK or your own server hand-off.
- Test by *tapping* (Notes, Messages) or `xcrun simctl openurl booted <url>`; see Apple's copy at `app-site-association.cdn-apple.com/a/v1/<domain>`.